Hamiltonian

I am considering the interaction term H_I of a Hamiltonian of a system of N particles with positions $\{\bar{\mathbf{R}}_l\}_{l\in\{1,\dots,N\}}$. It shall consist of two-body interactions, which depend on the position of the involved particles. Thus H_I can be written in the form

$$H_I(\{\bar{\mathbf{R}}_l\}) = \sum_{i,j;\, i\neq j} V_{i,j}(\bar{\mathbf{R}}_i, \bar{\mathbf{R}}_j)$$

In order to being able to derive the second order expansion of H_I, that will be needed later, in a more formal way, I write the two-body interactions as dependend on all positions.

$$H_I(\{\bar{\mathbf{R}}_l\}) = \sum_{i,j;\, i\neq j} V_{i,j}(\{\bar{\mathbf{R}}_l\})$$

The $V_{i,j}$ are formally functions of all positions $\{\bar{\mathbf{R}}_l\}$, i.e $V_{i,j}(\{\bar{\mathbf{R}}_l\})$. But of course they only depend on $\bar{\mathbf{R}}_i$ and $\bar{\mathbf{R}}_j$, i.e.

$$\frac{\partial V_{i,j}}{\partial \bar{\mathbf{R}}_l} = 0 \quad \text{if } i \neq l \neq j$$

Now I expand the interaction Hamiltonian in small displacements $\{\delta\mathbf{R}_l\}$ from the tweezer centers $\{\mathbf{R}_l\}$.

$$\begin{aligned}
H_I(\{\mathbf{R}_l + \delta\mathbf{R}_l\}) &= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n,m} \delta\mathbf{R}_n\,\frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n\partial\mathbf{R}_m}\,\delta\mathbf{R}_m \\
&= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n\neq m} \delta\mathbf{R}_n\,\frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n\partial\mathbf{R}_m}\,\delta\mathbf{R}_m + \frac{1}{2}\sum_{n=m} \delta\mathbf{R}_n\,\frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n\partial\mathbf{R}_m}\,\delta\mathbf{R}_m \\
&= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n\neq m}\sum_{i,j;\, i\neq j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n\partial\mathbf{R}_m}\,\delta\mathbf{R}_m \\
&+ \frac{1}{2}\sum_n\sum_{i,j;\, i\neq j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n^2}\,\delta\mathbf{R}_n \\
&= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n\neq m}\sum_{i,j;\, i\neq j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n\partial\mathbf{R}_m}\,\delta\mathbf{R}_m\,(\delta_{n=i,m=j} + \delta_{n=j,m=i}) \\
&+ \frac{1}{2}\sum_n\sum_{i,j;\, i\neq j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n^2}\,\delta\mathbf{R}_n\,(\delta_{n=i} + \delta_{n=j})
\end{aligned}$$

Using the fact, that

$$\delta\mathbf{R}_i\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_i\partial\mathbf{R}_j}\,\delta\mathbf{R}_j = \delta\mathbf{R}_j\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_j\partial\mathbf{R}_i}\,\delta\mathbf{R}_i$$

the expansion can be written as

$$\begin{aligned}
H_I(\{\mathbf{R}_l + \delta\mathbf{R}_l\}) = H_I(\{\mathbf{R}_l\}) &+ \sum_i \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i}\,\delta\mathbf{R}_i + \sum_{i,j;\, i\neq j} \delta\mathbf{R}_i\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_i\partial\mathbf{R}_j}\,\delta\mathbf{R}_j \\
&+ \sum_{i,j;\, i\neq j} \delta\mathbf{R}_i\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_i^2}\,\delta\mathbf{R}_i
\end{aligned}$$

If I introduce a matrix U the expanded Hamiltonian can be brought into the same form as in the paper.

$$H_I(\{\mathbf{R}_l + \delta\mathbf{R}_l\}) = H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \sum_{n,m} \delta\mathbf{R}_n\,U_{n,m}\,\delta\mathbf{R}_m$$

with

$$U_{n,m} = \frac{\partial^2 V_{n,m}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \quad \text{if } n \neq m$$
$$U_{n,n} = \sum_{j;\, j\neq n} \frac{\partial^2 V_{n,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2}$$

Swithing again to the reduced writting, where V_{ij} is only writen as a function of the variables of which it actually depends, reads

$$U_{n,m} = \frac{\partial^2 V_{n,m}(\mathbf{R}_n, \mathbf{R}_m)}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \quad \text{if } n \neq m$$
$$U_{n,n} = \sum_{j;\, j\neq n} \frac{\partial^2 V_{n,j}(\mathbf{R}_n, \mathbf{R}_j)}{\partial \mathbf{R}_n^2}$$

This result is a good point to further specify the potential that is used in the next sections. The potential will only depend on the relative vector between the two positions, i.e. $\mathbf{r}_{ij} = \mathbf{R}_i - \mathbf{R}_j$. Using the derivative

$$\frac{\partial \mathbf{r}_{ij}}{\partial \mathbf{R}_i} = \mathbb{1} = -\frac{\partial \mathbf{r}_{ij}}{\partial \mathbf{R}_j}$$

the derivatives of the U matrix can be written as

$$\begin{aligned}
\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} &= \frac{\partial}{\partial \mathbf{R}_n} \left(\frac{\partial V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \right) \\
&= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_n} + \frac{\partial V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}} \frac{\partial^2 \mathbf{r}_{nm}}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \\
&= -\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2}
\end{aligned}$$

and

$$\begin{aligned}
\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{R}_m^2 m} &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \frac{\partial \mathbf{r}_{nm}}{\partial \mathbf{R}_m} \\
&= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2}
\end{aligned}$$

Thus

$$U_{n,m} = -\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \quad \text{if } n \neq m$$
$$U_{n,n} = \sum_{j;\, j\neq n} \frac{\partial^2 V_{n,j}(\mathbf{r}_{nj})}{\partial \mathbf{r}_{nj}^2}$$

At this point I can eliminate the linear term of the Hamiltonian by introducing the trap potential

$$H_0(\{\delta \mathbf{R}_l\}) = \sum_n \frac{1}{2} m\omega^2 \, \delta \mathbf{R}_n^2$$

and redefining the trap centers by a equal shift of all positions $\{\delta \mathbf{R}_l\} = \{\delta \tilde{\mathbf{R}}_l + \boldsymbol{\Delta}\}$. Neglection all constant terms the Hamiltonian up to second order reads

$$\begin{aligned}
H(\{\delta \mathbf{R}_l\}) &= \sum_n \frac{1}{2} m\omega^2 \, (\delta \tilde{\mathbf{R}}_n + \boldsymbol{\Delta}_n)^2 + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} (\delta \tilde{\mathbf{R}}_n + \boldsymbol{\Delta}_n) + \sum_{n,m} (\delta \tilde{\mathbf{R}}_n + \boldsymbol{\Delta}_n)\, U_{n,m}\, (\delta \tilde{\mathbf{R}}_m + \boldsymbol{\Delta}_m) \\
&= \sum_n \frac{1}{2} m\omega^2 \, \delta \tilde{\mathbf{R}}_n^2 + \sum_{n,m} \delta \tilde{\mathbf{R}}_n \, U_{n,m}\, \delta \tilde{\mathbf{R}}_m + \sum_n \left(\frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} + \left(2 \sum_m \delta \tilde{\mathbf{R}}_m \, U_{m,n} + m\omega^2 \right) \boldsymbol{\Delta}_n \right) \delta \tilde{\mathbf{R}}_n,
\end{aligned}$$

where it was used that U is symmetric. The linear term vanishes, if $\mathbf{\Delta}_n$ is chosen such that

$$\mathbf{\Delta}_n = -\left(2\sum_m \delta\tilde{\mathbf{R}}_m\, U_{m,n} + m\omega^2\right)^{-1} \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}$$

Thus

$$H(\{\delta\tilde{\mathbf{R}}_l\}) = \sum_n \frac{1}{2} m\omega^2\, \delta\tilde{\mathbf{R}}_n^2 + \sum_{n,m} \delta\tilde{\mathbf{R}}_n\, U_{n,m}\, \delta\tilde{\mathbf{R}}_m$$

To describe the deviation from the center in a quantum mechanical way, the displacements are now quantized as phonons using the quadrature operators

$$\delta\tilde{\mathbf{R}}_n = \frac{1}{\sqrt{m\omega}}\,(a_n + a_n^\dagger)$$

If we exploit that the interactions realized in U are much smaller compared to the trap frequency, i.e. $U_{n,m} \ll m\omega^2$, the creation and annhiliation of phonons is supressed and the dynamics only involves phonon hopping, thus combinations aa or $a^\dagger a^\dagger$ are discarded.

$$\begin{aligned} H &= \sum_n \omega\, a_n^\dagger a_n + 2\sum_{n,m} \frac{U_{n,m}}{m\omega}\, a_n^\dagger a_m \\ &=: \sum_n \omega\, a_n^\dagger a_n + \sum_{n,m} \tilde{U}_{n,m}\, a_n^\dagger a_m \end{aligned}$$

Aligned Dipole-Dipole Interactions

In this section the derivative of the dipole interaction will be performed in the case where all dipoles point into the same direction. The subsripts lableling the different positions are droped in order to make the notation plainer.

$$V_{dd}(\mathbf{r}) = \frac{1}{r^3} - 3\,\frac{(\mathbf{r}\cdot\hat{\mathbf{d}})^2}{r^5},$$

with $\mathbf{r} = \mathbf{R}_i - \mathbf{R}_j$ beeing the distance between any to sites i and j $(i \neq j)$. The first derivative with respect to component α of $\mathbf{r}$, $\alpha \in \{x, y, z\}$ reads

$$\frac{\partial V_{dd}}{\partial r_\alpha} = -\frac{3r_\alpha}{r^5} + \frac{15 r_\alpha(\mathbf{r}\cdot\hat{\mathbf{d}})^2}{r^7} - \frac{6(\mathbf{r}\cdot\hat{\mathbf{d}})d_\alpha}{r^5},$$

the second derivative reads consequently

$$\begin{aligned} \frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= -3\,\frac{\delta_{\alpha\beta}}{r^5} + 15\,\frac{r_\alpha r_\beta}{r^7} + 15\,\frac{\delta_{\alpha\beta}(\mathbf{r}\cdot\hat{\mathbf{d}})^2}{r^7} - 15\cdot 7\,\frac{r_\alpha r_\beta(\mathbf{r}\cdot\hat{\mathbf{d}})^2}{r^9} \\ &\quad + 30\,\frac{d_\alpha r_\beta + d_\beta r_\alpha}{r^7}\,(\mathbf{r}\cdot\hat{\mathbf{d}}) - 6\,\frac{d_\alpha d_\beta}{r^5} \\ \\ &= \delta_{\alpha\beta}\left(-3\,\frac{1}{r^5} + 15\,\frac{(\mathbf{r}\cdot\hat{\mathbf{d}})^2}{r^7}\right) - 6\,\frac{d_\alpha d_\beta}{r^5} + 30\,\frac{d_\alpha r_\beta + d_\beta r_\alpha}{r^7}\,(\mathbf{r}\cdot\hat{\mathbf{d}}) \\ &\quad + 15\, r_\alpha r_\beta\left(\frac{1}{r^7} - 7\,\frac{(\mathbf{r}\cdot\hat{\mathbf{d}})^2}{r^9}\right). \end{aligned}$$

General Dipole-Dipole Interactions

In the case where not all dipole moments are aligned, the properties of the interaction get extended to

$$V_{dd}(\mathbf{r}) = \frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^3} - 3\,\frac{(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^5}.$$

The first derivative reads

$$\frac{\partial V_{dd}}{\partial r_\alpha} = -\frac{3\,(\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2)\,r_\alpha}{r^5} + \frac{15 r_\alpha(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^7} - \frac{3\,\Big((\mathbf{r}\cdot\hat{\mathbf{d}}_1)\,d_{2,\alpha} + (\mathbf{r}\cdot\hat{\mathbf{d}}_2)\,d_{1,\alpha}\Big)}{r^5},$$

the second derivative reads consequently

$$\begin{aligned}
\frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= -3\,\frac{\delta_{\alpha\beta}\,(\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2)}{r^5} + 15\,\frac{r_\alpha r_\beta\,(\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2)}{r^7} + 15\,\frac{\delta_{\alpha\beta}(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^7} - 15\cdot 7\,\frac{r_\alpha r_\beta(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^9}\\
&\quad + 15\left[\frac{d_{1,\alpha}r_\beta + d_{1,\beta}r_\alpha}{r^7}\,(\mathbf{r}\cdot\hat{\mathbf{d}}_2) + \frac{d_{2,\alpha}r_\beta + d_{2,\beta}r_\alpha}{r^7}\,(\mathbf{r}\cdot\hat{\mathbf{d}}_1)\right] - 3\,\frac{d_{1,\alpha}d_{2,\beta} + d_{2,\alpha}d_{1,\beta}}{r^5}\\
&= \delta_{\alpha\beta}\left(-3\,\frac{\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2}{r^5} + 15\,\frac{(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^7}\right) - 3\,\frac{d_{1,\alpha}d_{2,\beta} + d_{2,\alpha}d_{1,\beta}}{r^5}\\
&\quad + 15\left[\frac{d_{1,\alpha}r_\beta + d_{1,\beta}r_\alpha}{r^7}\,(\mathbf{r}\cdot\hat{\mathbf{d}}_2) + \frac{d_{2,\alpha}r_\beta + d_{2,\beta}r_\alpha}{r^7}\,(\mathbf{r}\cdot\hat{\mathbf{d}}_1)\right]\\
&\quad + 15\,r_\alpha r_\beta\left(\frac{\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2}{r^7} - 7\,\frac{(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^9}\right).
\end{aligned}$$

For the the Van der Waals interaction the potential has to be squared. Therefore the second derivative reads

$$\begin{aligned}
\frac{\partial V_{dd}^2}{\partial r_\alpha} &= 2\,V_{dd}\frac{\partial V_{dd}}{\partial r_\alpha}\\
\frac{\partial^2 V_{dd}^2}{\partial r_\alpha \partial r_\beta} &= 2\,\frac{\partial V_{dd}}{\partial r_\alpha}\frac{\partial V_{dd}}{\partial r_\beta} + 2\,V_{dd}\frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta}
\end{aligned}$$

Winding Number

A chiral Hamiltonian is defined through the property, that it anticommutes with a unitary hermitian operator Γ

$$\Gamma H \Gamma = -H \quad \text{with} \quad \Gamma^2 = 1. \tag{0.1}$$

From this property follows that the Hamiltonian can be written in block off-diagonal form

$$H = \begin{pmatrix} 0 & h \\ h^\dagger & 0 \end{pmatrix}, \tag{0.2}$$

where h is, in general, a non-Hermitian matrix. This structure also applies for the matrix in momentum space and allows the definition of a winding number, which characterizes the topological properties of the system. If the Hamiltonian is 2×2 dimensional, the winding number can be defined through the decomposition of H in terms of the Pauli matrices

$$H = \mathbf{n}\cdot\boldsymbol{\sigma}, \tag{0.3}$$

where $\mathbf{n} = (n_x, n_y, 0)$ is a vector in the xy-plane. The winding number is then given by the winding of the vector $\mathbf{n}$ around the origin when traversing the Brillouin zone. Identifying the xy-plane of $\mathbf{n}$ with the complex plane, which defines a path $\mathbf{n}(k) \to \gamma(k)$, the winding number can be calculated through

$$w = \frac{1}{2\pi i}\oint_{\gamma_k} \frac{1}{z}\,dz = \frac{1}{2\pi i}\int_{\mathrm{BZ}} dk \frac{1}{\gamma(k)}\frac{d\gamma(k)}{dk}.$$

In terms of the Hamiltonian above this reads

$$w = \frac{1}{2\pi i}\int_{\mathrm{BZ}} dk\,\frac{h'(k)}{h(k)}.$$

Generalizing the winding number to higher dimensional Hamiltonians is possible through the use of the block off-diagonal form. The winding number then reads [1, 2]

$$w = \frac{1}{2\pi i}\int_{\mathrm{BZ}} dk\,\mathrm{Tr}\left[h^{-1}(k)\partial_k h(k)\right].$$

Diagonalizing h as $h = S\Lambda S^{-1}$ with the diagonal matrix $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, ...)$ and the invertible matrix S, the trace can be rewritten as

$$\begin{aligned}
\mathrm{Tr}\left[h^{-1}\partial_k h\right] &= \mathrm{Tr}\left[S\Lambda^{-1}S^{-1}\partial_k(S\Lambda S^{-1})\right] \\
&= \mathrm{Tr}\left[\Lambda^{-1}\partial_k\Lambda + S^{-1}\partial_k S + (\partial_k S^{-1})S\right] \\
\mathrm{Tr}\left[S^{-1}\partial_k S + S\partial_k S^{-1}\right] &= \mathrm{Tr}\left[\partial_k\left(SS^{-1}\right)\right] = 0, \\
\Rightarrow \mathrm{Tr}\left[h^{-1}\partial_k h\right] &= \mathrm{Tr}\left[\Lambda^{-1}\partial_k\Lambda\right] = \sum_i \frac{\partial_k\lambda_i}{\lambda_i} = \partial_k\left(\sum_i \ln(\lambda_i)\right) = \partial_k \ln(\det[h]).
\end{aligned}$$

Another useful definition of the winding number is through the so-called Q-matrix, which is defined through the projectors onto the eigenstates of the Hamiltonian. As the Hamiltonian is chiral, the eigenvalues come in pairs $\pm E$. We introduce a labeling of the eigenvalues, such that $E_j = -E_{-j}$ with $j = 1, 2, ..., N$ and N being half the dimension of the Hamiltonian. The corresponding eigenstates are labeled accordingly ($|\psi_j\rangle$) and posses the property $\Gamma|\psi_j\rangle = |\psi_{-j}\rangle$. The Q-matrix is defined as

$$Q = \sum_{j=1}^{N}\left(|\psi_j\rangle\langle\psi_j| - |\psi_{-j}\rangle\langle\psi_{-j}|\right).$$

As a consequence of the chiral symmetry, the Q-matrix anticommutes with Γ and can therefore be written in block off-diagonal form in the basis, where Γ is diagonal, which is also the case in which H is block off-diagonal.

$$\begin{aligned}
\Gamma Q\Gamma &= \sum_{j=1}^{N}\left(|\psi_{-j}\rangle\langle\psi_{-j}| - |\psi_j\rangle\langle\psi_j|\right) = -Q \\
\Rightarrow Q &= \begin{pmatrix} 0 & q \\ q^\dagger & 0 \end{pmatrix}
\end{aligned}$$

The fact that the winding number of the q-matrix is the same as that of h can be shown with the help of the following two properties.

$$\begin{aligned}
Q^2 = 1 \quad &\Rightarrow \quad qq^\dagger = \mathbb{1}, \quad q^\dagger q = \mathbb{1} \\
QHQ = H \quad &\Rightarrow \quad qh^\dagger q = h^\dagger
\end{aligned}$$

From the second property follows that $q^\dagger h = h^\dagger q$. Thus $h^\dagger q$ is a hermitian matrix and therefore has no winding number, since the eigenvalues are real. Hence

$$\begin{aligned}
0 &= \mathrm{Tr}\left[q^\dagger(h^\dagger)^{-1}\partial_k(h^\dagger q)\right] \\
&= \mathrm{Tr}\left[q\,q^\dagger\,(h^\dagger)^{-1}\partial_k h^\dagger + h^\dagger(h^\dagger)^{-1}\,q^\dagger\partial_k q\right] \\
\Rightarrow \mathrm{Tr}\left[q^{-1}\partial_k q\right] &= -\mathrm{Tr}\left[(h^\dagger)^{-1}\partial_k h^\dagger\right] \\
&= \mathrm{Tr}\left[h^{-1}\partial_k h\right]
\end{aligned}$$

There is another useful property, for which we consider the basis in which $\Gamma = \sigma_z \otimes \mathbb{1}_N$. This is the same diagonal basis as previously used. In this basis we can represent all eigenvalues of H as a stack of to N-component vectors. The eigenvector for negative energies is written as

$$|\psi_i^-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}|h_{1i}\rangle \\ |h_{2i}\rangle\end{pmatrix} \qquad i \in \{1, \dots, N\}$$

Because of the chirality of the Hamiltonian the eigenvector with positive energy is already determined by this choice and reads

$$|\psi_i^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} |h_{1i}\rangle \\ -|h_{2i}\rangle \end{pmatrix}$$

The matrix Q is represented in the following form

$$\begin{aligned} Q &= \sum_i \begin{pmatrix} 0 & |h_{1i}\rangle \langle h_{2i}| \\ |h_{2i}\rangle \langle h_{1i}| & 0 \end{pmatrix} \\ \Rightarrow \quad q &= \sum_i |h_{1i}\rangle \langle h_{2i}| \end{aligned}$$

If we now compute the kernel of the winding number with respect to q we get

$$\begin{aligned} \mathrm{Tr}[q^\dagger \partial_k q] &= \mathrm{Tr}\left[\sum_{i,j} |h_{2i}\rangle \langle h_{1i}| \, \partial_k \, (|h_{1j}\rangle \langle h_{2j}|)\right] \\ &= \sum_{i,j} \mathrm{Tr}\left[|h_{2i}\rangle \langle h_{1i}| \, (\partial_k |h_{1j}\rangle) \, \langle h_{2j}|\right] + \sum_{i,j} \mathrm{Tr}\left[|h_{2i}\rangle \langle h_{1i}| \, |h_{1j}\rangle \, (\partial_k \langle h_{2j}|)\right] \\ &= \sum_i \langle h_{1i}| \, (\partial_k |h_{1i}\rangle) + \sum_i (\partial_k \langle h_{2i}|) \, |h_{2i}\rangle \end{aligned}$$

When computing the winding number the second term contributes with a minus sign compared to its complex conjugated self, i.e.

$$\int_{\mathrm{BZ}} dk \; (\partial_k \langle h_{2i}|) \, |h_{2i}\rangle = - \int_{\mathrm{BZ}} dk \; \langle h_{2i}| \, (\partial_k |h_{2i}\rangle)$$

Thus the winding number reads

$$\begin{aligned} w &= \frac{1}{2\pi i} \int_{\mathrm{BZ}} dk \, \mathrm{Tr}[q^\dagger \partial_k q] \\ &= \frac{1}{2\pi i} \sum_i \int_{\mathrm{BZ}} dk \; (\langle h_{1i}| \, \partial_k \, |h_{1i}\rangle - \langle h_{2i}| \, \partial_k \, |h_{2i}\rangle) \end{aligned}$$

This can be expressed with respect to the full eigenvectors of the Hamiltonian via

$$\begin{aligned} w &= \frac{1}{\pi i} \sum_i \int_{\mathrm{BZ}} dk \; \langle \Gamma \psi_i^- | \, \partial_k \, |\psi_i^-\rangle \\ &= \frac{1}{\pi i} \sum_i \int_{\mathrm{BZ}} dk \; \langle \psi_i^+ | \, \partial_k \, |\psi_i^-\rangle \\ &= \frac{1}{2\pi i} \sum_i \int_{\mathrm{BZ}} dk \; (\langle h_{1i}| \, \partial_k \, |h_{1i}\rangle - \langle h_{2i}| \, \partial_k \, |h_{2i}\rangle) \end{aligned}$$

The quantity $\sum_i \langle \Gamma \psi_i^- | \, \partial_k \, |\psi_i^-\rangle$ can be related to a polarisation [3].

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